✅ SOP #14 – ATI (Analysis to Improve) for Sump Pump Systems

Goal: Optimize stormwater, wastewater, and process water removal while minimizing electrical and mechanical wear

📊 1. Efficiency Targets

Metric	Ideal Benchmark	Optimization Strategy
Pump Runtime Efficiency	>80% within target range	Float tuning, avoid short cycling
Power per Gallon Pumped	As low as feasible	Optimize discharge height and routing
Backup System Availability	100% operational	Monthly battery/auxiliary check
Preventive Maintenance Score	100% per month	Use logs and alarms to enforce schedule

⚙️ 2. Improvement Methods

Dual Pump with Alternation Relay: Extend lifespan through load sharing
High-Water Alarm with SMS/Email Alerts: Improve responsiveness
Install Vibration Switches or Sensors: Monitor mechanical degradation
Backflow Valve Upgrades: Improve flow efficiency and prevent reentry
Pit Cleaning and Sludge Monitoring: Prevent solids buildup that strains pumps

⚖️ 3. Codes & Standards

NFPA 70 (NEC): Electrical wiring and control compliance
OSHA 1910.305: Electrical and guarding requirements
EPA Industrial Discharge Permits: For certain facilities
IFC/IBC: Flood protection and pump redundancy
Inspection Logs: Retain for 3+ years as best practice